



CO3-AT1-Debugging and Optimization

[← Back](#)

QUESTIONS

Q1ATM Withdrawal – Exception
Handling Debugging
10 marks**Q2**Hospital Registration –
NullPointerException
Debugging
10 marks**Q3**E-Commerce Product Search –
Array Exception Debugging
10 marks**Q4**Railway Reservation –
Synchronization Debugging
10 marks**Q5**Bank Account – Race Condition
Debugging

Q3 E-Commerce Product Search – Array Exception Debugging

10 marks

An online shopping application allows customers to select a product using a product number from 1 to 5.

The following program contains an array indexing error.

Task:

1. Identify the error.
2. Correct the product-number and array-index logic.
3. Handle invalid product numbers.
4. Handle non-numeric input.
5. Submit the COMPLETE corrected Java program.
6. The program must display "Search completed" after processing.

Faulty Program:

```
import java.util.Scanner;

class ProductSearch {

    public static void main(String[] args) {

        Scanner sc = new Scanner(System.in);
```

10 marks

5/5 answered

```
String[] products = {  
    "Laptop",  
    "Mobile",  
    "Tablet",  
    "Headphone",  
    "Keyboard"  
};  
  
System.out.print("Enter product number (1-5): ");  
int productNumber = sc.nextInt();  
  
System.out.println("Selected product: "  
    + products[productNumber]);  
  
System.out.println("Search completed");  
}  
}
```

Correct the program so that product numbers 1, 2, 3, 4 and 5 correctly access the corresponding products.

🔧 **Debugging Task:** Find and fix the bugs in the code shown above. Write your corrected code in the editor below.

Your Fixed Code

```
1 import java.util.Scanner;  
2 class ProductSearch{  
3     public static void main(String[] args) {  
4         Scanner sc = new Scanner(System.in);  
5         String[] products={
```

Test Input

1

```
6         "Laptop",
7         "Mobile",
8         "Tablet",
9         "Headphone",
10        "Keyboard"
11    };
12    int n = sc.nextInt();
13    System.out.println("Selected Product :"+ pr
14    System.out.println("Search Completed");
15    }
16 }
```

Output

Selected Product :Laptop
Search Completed

 Pending manual review

 Submitted

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